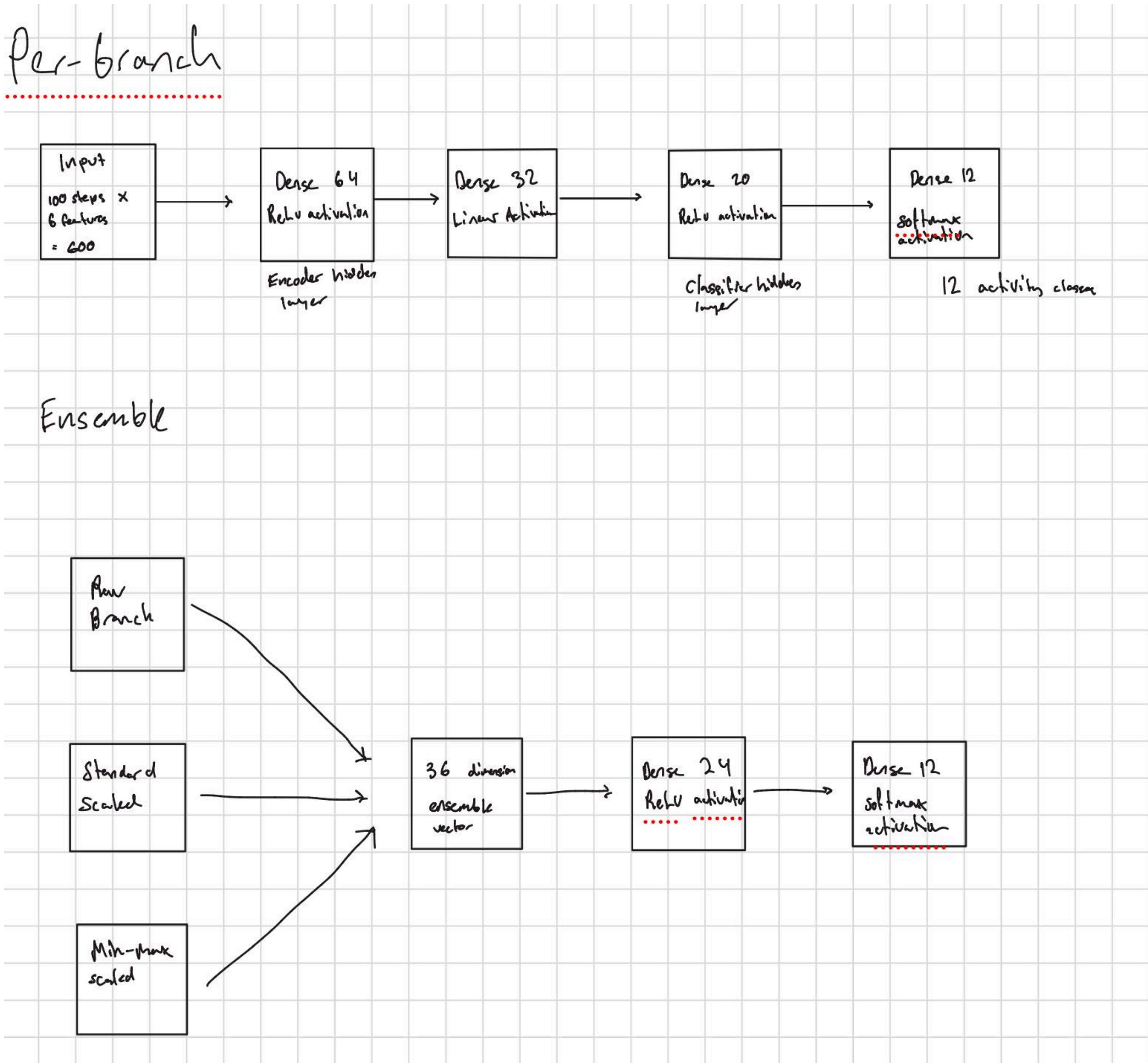
Part 1:
https://studio.edgeimpulse.com/public/1080769/live

```
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 14 ms, inference 6 ms, anomaly 0 ms, postprocessing 49 us
#Classification predictions:
  idle: 0.996094
  lift: 0.000000
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 14 ms, inference 6 ms, anomaly 0 ms, postprocessing 49 us
#Classification predictions:
  idle: 0.996094
  lift: 0.000000
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 14 ms, inference 6 ms, anomaly 0 ms, postprocessing 49 us
#Classification predictions:
  idle: 0.000000
  lift: 0.996094
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 14 ms, inference 6 ms, anomaly 0 ms, postprocessing 67 us
#Classification predictions:
  idle: 0.000000
  lift: 0.996094
```

Part 2:

Question 1:



Question 2: Pruning is applied by using magnitude based pruning to identify the weights with the smallest value in each layer, a sparsity schedule of 10% to 80% is used. We must strip before doing the QAT to turn the wrapper layers into regular dense layers. QAT wraps layers with another wrapper to insert fake nodes, and if we used that with the pruning wrapper it would cause conflicts. QAT is applied by inserting simulated quantization ops into the stripped model and then fine tuning the weights so that they adapt to the precision loss that will come at int8 inference. We need an extra mask-enforcement step because it ensures that the weights that we pruned originally stay pruned and are not changed by the QAT, and we can see in our case the sparsity stayed at 79.4% before and after.The model is converted to int8 using TFLite converter which forces both the input and output to int8. A representative dataset is required for full integer quantization because there needs to be a real activation range at each layer to compute the scale and zero point for int8 mapping. This is important for our board because it

has limited memory, and the pruning and quantization helped to lower the models footprint significantly without affecting the accuracy much.

Question 3:

```
Output    Serial Monitor ×
Message (Enter to send message to 'Arduino Nano 33 BLE' on 'COM4')
Class index: 2
Confidence score: 0.6523
----------------------------------------
Samples collected: 25
Samples collected: 50
Samples collected: 75
Samples collected: 100
Window collected. Running ensemble inference...
Raw model scores: 0.0000, 0.0000, 0.9219, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0742, 0.0039
Standard-scaled model scores: 0.0000, 0.2969, 0.6602, 0.0000, 0.0430, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000
Min-max-scaled model scores: 0.0000, 0.0234, 0.3711, 0.0000, 0.1914, 0.1367, 0.2266, 0.0000, 0.0156, 0.0039, 0.0000, 0.0391
Meta-classifier scores: 0.0039, 0.0234, 0.6719, 0.0000, 0.0000, 0.0078, 0.0000, 0.0820, 0.2070, 0.0000, 0.0000, 0.0039
----------------------------------------
Final prediction: Lying down
Class index: 2
Confidence score: 0.6719
----------------------------------------
Samples collected: 25
Samples collected: 50
Samples collected: 75
Samples collected: 100
Window collected. Running ensemble inference...
Raw model scores: 0.0000, 0.0000, 0.9297, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0664, 0.0000
Standard-scaled model scores: 0.0000, 0.3203, 0.6445, 0.0000, 0.0352, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000
Min-max-scaled model scores: 0.0000, 0.0234, 0.3672, 0.0000, 0.1875, 0.1367, 0.2227, 0.0000, 0.0234, 0.0000, 0.0000, 0.0391
Meta-classifier scores: 0.0039, 0.0273, 0.6680, 0.0000, 0.0000, 0.0078, 0.0000, 0.0820, 0.2070, 0.0000, 0.0000, 0.0039
----------------------------------------
Final prediction: Lying down
Class index: 2
Confidence score: 0.6680
----------------------------------------
```

I think that the output is acceptable, however it had low confidence in most of the predictions. It maintained a similar confidence level for all of the “lying down” classifications. However it never has a higher confidence than 67% because of the high percentages for other activities, because it is integrating branches that all disagree with each other. It is not unstable, but not confident either. One practical change could be to use a moving average over a few windows before printing a final label; this could help prevent a single noisy window from changing the label when the board moves.

Question 4:

One unexpected behavior I observed was that the board would sometimes classify as “lying down” when I moved the board around. This could happen because the training data from mHealth was probably recorded on different hardware and the sensors were in different places on a person than my board was. It could also be because the body motions from these activities are different than any way I moved my board.